

Math 526, F09
Quiz 6

Name: *Answer*
SSN: xxx-xx-

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (20 points) Let $A = \begin{bmatrix} 1 & 3 & 1 & 2 \\ 2 & 6 & 4 & 8 \\ -2 & -6 & 0 & 0 \end{bmatrix}$ and $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$.

(a) Reduce $[A \ b]$ to $[R \ d]$ (the reduced echelon form) to find the condition on b_1, b_2, b_3 such that $Ax = b$ is solvable.

$$\text{Solution: } [A \ b] = \left[\begin{array}{cccc|c} 1 & 3 & 1 & 2 & b_1 \\ 2 & 6 & 4 & 8 & b_2 \\ -2 & -6 & 0 & 0 & b_3 \end{array} \right] \xrightarrow{\substack{R_2 - 2R_1 \\ R_3 + 2R_1}} \left[\begin{array}{cccc|c} 1 & 3 & 1 & 2 & b_1 \\ 0 & 0 & 2 & 4 & b_2 - 2b_1 \\ 0 & 0 & 2 & 4 & b_3 + 2b_1 \end{array} \right]$$

$$\xrightarrow{R_3 - R_2} \left[\begin{array}{cccc|c} 1 & 3 & 1 & 2 & b_1 \\ 0 & 0 & 2 & 4 & b_2 - 2b_1 \\ 0 & 0 & 0 & 0 & b_3 - b_2 + 4b_1 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{cccc|c} 1 & 3 & 1 & 2 & b_1 \\ 0 & 0 & 1 & 2 & \frac{b_2}{2} - b_1 \\ 0 & 0 & 0 & 0 & b_3 - b_2 + 4b_1 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{cccc|c} 1 & 3 & 0 & 0 & 2b_1 - \frac{b_2}{2} \\ 0 & 0 & 1 & 2 & \frac{b_2}{2} - b_1 \\ 0 & 0 & 0 & 0 & b_3 - b_2 + 4b_1 \end{array} \right] = [R \ d]$$

$$Ax = b \text{ is solvable} \Leftrightarrow b_3 - b_2 + 4b_1 = 0$$

(b) Find the complete solution to $Ax = b$ with $b = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}$.

$$b = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix} \Rightarrow d = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ 0 \end{bmatrix} \quad R = \begin{bmatrix} 1 & 3 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$x = x_p + x_n \rightarrow$ special solutions
 $\rightarrow$ particular solution

Pivot variables: x_1, x_3

free variables: x_2, x_4 .

i) Particular solution: $Ax_p = b \Leftrightarrow Rx_p = d$.

$$\begin{cases} x_1 + 3x_2 = \frac{1}{2} \\ x_3 + 2x_4 = \frac{1}{2} \end{cases}$$

$$\text{Set } x_2 = x_4 = 0 \Rightarrow x_1 = \frac{1}{2}, x_3 = \frac{1}{2}$$

$$\Rightarrow x_p = \begin{bmatrix} \frac{1}{2} \\ 0 \\ \frac{1}{2} \\ 0 \end{bmatrix}$$

ii) Special solutions: $Ax_n = 0 \Leftrightarrow Rx_n = 0$

$$\begin{cases} x_1 + 3x_2 = 0 \\ x_3 + 2x_4 = 0 \end{cases}$$

$$\text{Set } x_2 = 1, x_4 = 0 \Rightarrow x_1 = -3, x_3 = 0 \Rightarrow s_1 = \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{Set } x_2 = 0, x_4 = 1 \Rightarrow x_1 = 0, x_3 = -2 \Rightarrow s_2 = \begin{bmatrix} 0 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

Thus, $x = x_p + C_1 s_1 + C_2 s_2$

$$= \begin{bmatrix} \frac{1}{2} \\ 0 \\ \frac{1}{2} \\ 0 \end{bmatrix} + C_1 \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + C_2 \begin{bmatrix} 0 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$